Do Now · retrieval

Lesson 156 · Client-Side versus Server-Side Processing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why does PageRank divide a page's score among outgoing links?

Q2

What data structure follows LIFO?

Q3

What is a database foreign key?

Q4

What is the role of a router?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Browser tasks — JavaScript can update local interface promptly.

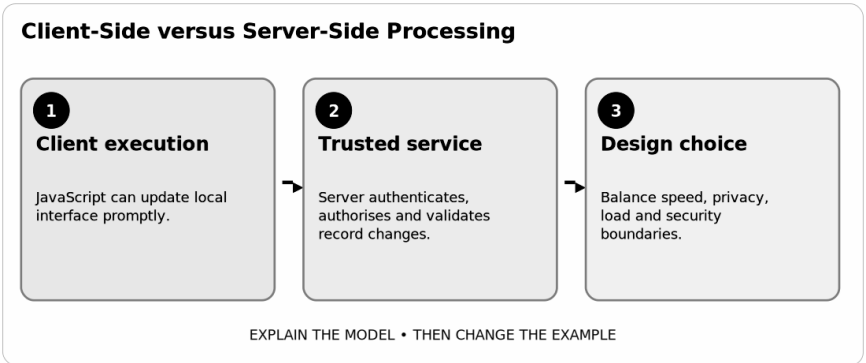
Core idea 2

Backend tasks — Server authenticates, authorises and validates record changes.

Core idea 3

Compare deployment — Balance speed, privacy, load and security boundaries.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

client-side processing — Computing performed in browser/device.

server-side processing — Computing performed by backend.

authentication — Checking claimed identity.

authorisation — Checking what a user may do.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A pupil edits a teacher-assigned mark in browser developer tools before submitting a form. How is the system protected?

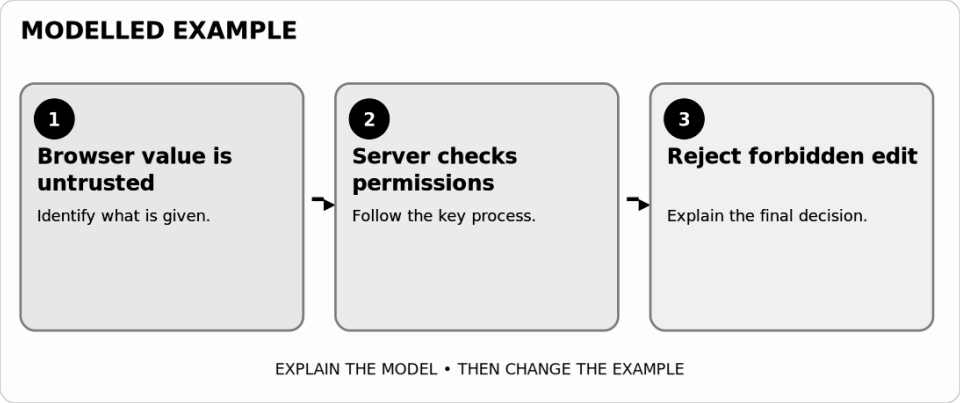
Follow the worked method

1. Browser value is untrusted — Identify the input or diagram and predict the first result.

2. Server checks permissions — Follow the second stage and explain its importance.

3. Reject forbidden edit — The server must authenticate the user and enforce edit permissions, ignoring unauthorised mark values. It validates requests and writes to the authoritative database only after checking access.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Public quiz
Identify the input or diagram and predict the first result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give two benefits and one limitation of client-side processing. [6]

Question 2

Explain why permission to change assessment results must be checked server-side. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A hub offers instant quiz hints, private reports and offline revision. Assign client/server roles and justify. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Browser validation is sufficient to authorise database changes.

Use key vocabulary from memory

1. Explain client-side processing in your own words.

2. Explain server-side processing in your own words.

3. Explain authentication in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.